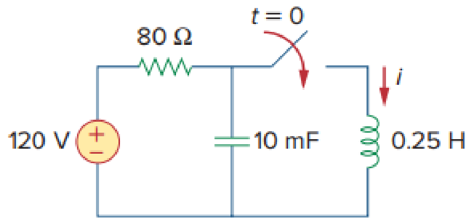
Chapter 8, Problem 53P

Problem

After being open for a day, the switch in the circuit of Fig. 8.99 is closed at $t = 0$. Find the differential equation describing $i(t)$, $t > 0$.

Figure 8.99:



Step-by-step solution

Step 1 of 3

Refer to circuit diagram in Figure 8.99 in the textbook.

At $t > 0$ the switch in the circuit is closed. The circuit diagram at $t > 0$ is shown in Figure 1.

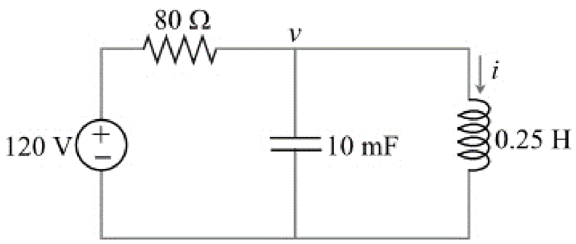


Figure 1

Step 2 of 3

The voltage across the inductor is,

$$\begin{aligned} v_L &= L\frac{di}{dt} \\ &= 0.25\frac{di}{dt} \end{aligned}$$

In the circuit shown in Figure 1, capacitor and inductor are connected in parallel.

It is known that, in parallel connection the voltage across each element is same. Therefore,

$$\begin{aligned} v &= v_L \\ &= 0.25\frac{di}{dt} \end{aligned}$$

Step 3 of 3

Apply the Kirchhoff's current law at node v.

$$\frac{v-120}{80}+\left(10\times10^{-3}\right)\frac{dv}{dt}+i=0$$

Substitute $0.25\left(\frac{di}{dt}\right)$ for v.

$$\frac{0.25\frac{di}{dt}-120}{80}+\left(10\times10^{-3}\right)\frac{d}{dt}\left(0.25\frac{di}{dt}\right)+i=0$$

$$0.25\frac{di}{dt}-120+(80)(0.25)\left(10\times10^{-3}\right)\frac{d^2i}{dt^2}+80i=0$$

$$0.2\frac{d^2i}{dt^2}+0.25\frac{di}{dt}-120+80i=0$$

$$\frac{d^2i}{dt^2}+\frac{0.25}{0.2}\frac{di}{dt}+\frac{80}{0.2}i=\frac{120}{0.2}$$

$$\frac{d^2i}{dt^2}+1.25\frac{di}{dt}+400i=600$$

Therefore, the differential equation for $i(t)$ is $\boxed{\frac{d^2i}{dt^2}+1.25\frac{di}{dt}+400i=600}$.